Laser measurement prediction with LSTM

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***Abstract*—This study investigates time series forecasting on a laser measurement dataset using Long Short-Term Memory (LSTM). The task is to predict one-step-ahead values and extend predictions recursively for future time steps. The data is normalized before training and rescaled for evaluation. A sliding window approach is used to construct input sequences, with sequence length treated as a tunable parameter. Model performance is assessed using Mean Absolute Error (MAE) and Mean Squared Error (MSE), along with visuals.**

***Index Terms*—LSTM, Deep Learning, Machine Learning, Prediction Model, Recursive Prediction.**

I. Introduction

Time series forecasting is widely used in machine learning and deep learning for domains such as finance, weather prediction, signal processing, and physical measurement data. The goal is to model temporal dependencies in sequential data in order to predict future values.

However, traditional machine learning methods often struggle in capturing long- term dependencies in time series data. Deep learning models, particularly recurrent architectures such as LSTMs, are well suited for learning temporal patterns in sequential data.

In this assignment, we implement LSTM models for one-step-ahead prediction. Additionally, we evaluate its performance in recursive multi-step forecasting and analyze the effect of sequence length on model performance.

Although other sequence models could be considered, this report focuses on an LSTM because it is suitable for sequential data and can be trained effectively on relatively small datasets. A small prior experiment showed little promise in the performance of other models compared to the LSTM.

II. LSTM

The Long Short-Term Memory (LSTM) network is a recurrent neural network type designed to capture temporal dependencies in sequential data. It uses gating mechanisms for controlling the information flow, which allows the model to retain long term dependencies and avoid the vanishing gradient problem. LSTMs are a suitable choice for this task because they can model temporal dependencies without requiring very large datasets.

III. Approach

A. Data Preprocessing

The dataset was loaded from a .mat file and then converted into one dimensional array. The data was normalized using mean and standard deviation to ensure stable training process and faster convergence.

B. Sequence Creation

A sliding window approach was implemented to construct training samples. Given a sequence length variable (seq_len), each input sample consisted of the previous seq_len values, while the target was the next value in the sequence.

C. Hyperparameter Tuning

The sequence length (seq_len) determines how many past time steps the model receives as input when predicting the next value. This is the hyperparameter that was tuned; Very short windows may miss relevant temporal patterns, while very long windows may increase model complexity and reduce recursive forecasting stability. The following values were tested:

- 10, 20, 30, 50, 80

The optimal sequence length was chosen based on the lowest Mean Squared Error. This was evaluated by training the model on the first available 800 datapoints and forecast the remaining 200. These forecasted values were compared against the true values. Results are shown in Table I.

TABLE I
Recursive forecast error by sequence length

Sequence Length	MAE	MSE
10	23.53	1824.12
20	30.92	1982.13
30	39.41	3706.01
50	38.12	3566.06
80	32.35	1828.07

A sequence length of 10 achieved the lowest MAE and MSE and was selected for the final model. Fig. 1 shows the recursive

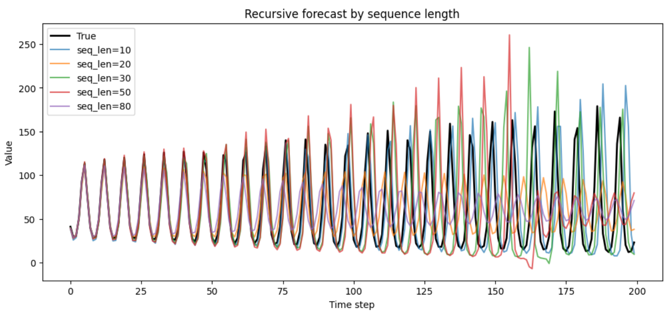

Fig. 1. Recursive 200-step forecast for different sequence lengths compared against true values. Shorter windows produce more stable recursive predictions.

forecasts for all tested values overlaid on the true data.

D. LSTM Architecture

Two stacked LSTM layers (64 and 32 units) followed by a single (Dense) output layer. The input is shape (seq_len,1) where seq_len = 10.The first LSTM layer uses `return_sequences=True` to pass its full output sequence to the second LSTM layer, which returns only its final hidden state.

E. Training Setup

The LSTM model was trained using the Adam optimizer, learning rate of 0.001 and MSE as the loss function. Training was run for 100 epochs with a batch size of 32. An 80/20 train-validation split was used to monitor for overfitting. The training and validation loss curves are shown in Fig. 2.

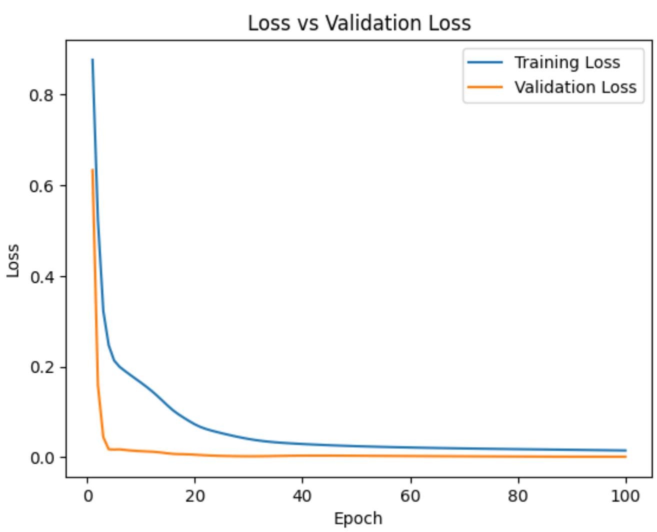

Fig. 2. Training and validation loss over 100 epochs for the LSTM model with seq_len = 10. Both curves converge without a large gap, suggesting limited overfitting.

F. Recursive Forecasting

The model takes the 10 last known values, predicts the next one, appends that prediction to the window (dropping oldest, keeping 10), and repeats 200 times. Future steps rely on previous (predictions) ones, meaning that errors can accumulate. Predictions are then rescaled from normalized values back to the original using stored mean and standard deviation.

IV. Results

A. Quantitative Results

Table I summarizes the recursive forecast error for each tested sequence length. The LSTM model with seq_len = 10 achieved the best performance with an MAE of 23.53 and MSE of 1824.12.

The model closely tracked true values across the 1000 available data plots. This is shown in one-step ahead prediction in Fig. 3.

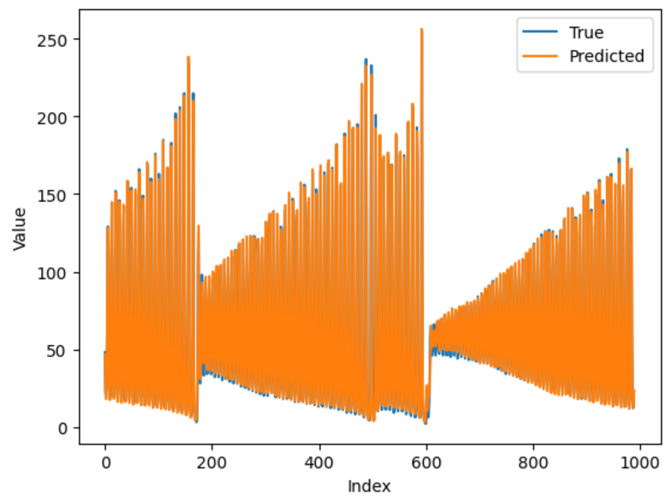

Fig. 3. One-step-ahead predictions versus true values across the full training dataset.

B. Qualitative Analysis

We can see that one-step ahead predictions closely match true values, which captures both the oscillations and build-up (and collapse) pattern of the laser intensity.

In the recursive 200-step forecast (Fig. 4) the seemingly realistic pattern is maintained. Though, small deviations become visible over longer horizon, most likely due to error accumulation.

C. Test Set Evaluation

After selecting a sequence length of 10, the final LSTM model was trained on the full training set and used to recursively predict the 200 points in Xtest.

Fig. 5 shows that the model captures the repeating peak structure in the early part of the test sequence, but its predictions deviate over time (especially after the sudden collapse of the value). The model was unable to predict the sudden collapse of the value properly.

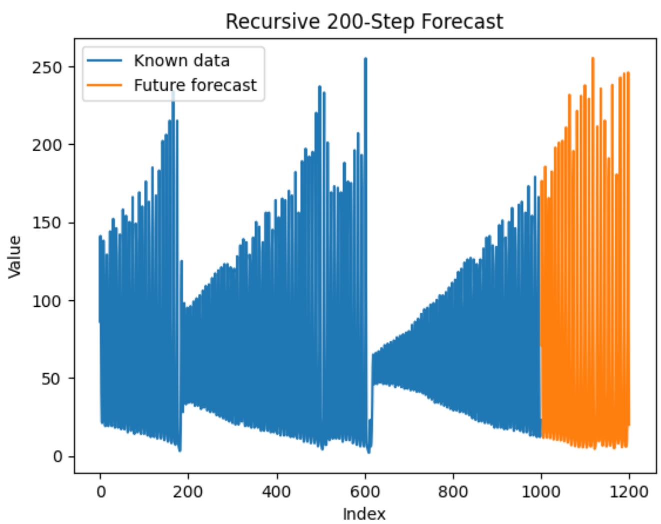

Fig. 4. Recursive 200-step forecast (orange) appended to the known training data (blue).

TABLE II
FINAL RECURSIVE TEST PERFORMANCE ON XTEST

Metric	Value
MAE	52.8240
MSE	5492.4248

V. DISCUSSION

The results show that the LSTM learned important structure in the laser measurement data. In the recursive forecast, the model preserves the strong oscillatory pattern, predicts approximately similar amplitudes, and follows the rough trend or envelope of the signal. This indicates that the model captured useful short-term dependencies from the training data.

However, the recursive test prediction also shows clear limitations. As the prediction horizon increases, small errors accumulate because every predicted value is reused as input for the next prediction. This causes phase drift: the model may still predict peaks of a reasonable size, but the peaks become increasingly misaligned with the real test data. Since the signal contains sharp oscillations, even small timing errors can lead to large MAE and MSE values.

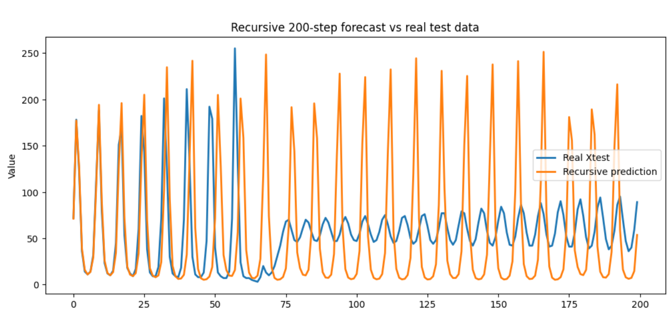

Fig. 5. Recursive 200-step LSTM forecast compared with the real Xtest values.

A possible explanation is that the laser measurement behaves like a sensitive dynamical system, where small deviations in earlier states can strongly affect later states. This is similar to the sensitivity observed in deterministic nonperiodic systems, as described by Lorenz [1]. Therefore, recursive forecasting is much harder than one-step-ahead prediction. The limited amount of training data also makes it difficult for the LSTM to learn all possible future regimes of the signal.

VI. CONCLUSION

This project used an LSTM model to predict laser measurement data. The model was trained for one-step-ahead prediction using normalized data and a sliding window approach. The sequence length was tuned by comparing recursive forecasts on the final part of the training data, and a sequence length of 10 gave the lowest validation MAE and MSE.

On the real Xtest data, the final model achieved an MAE of 52.8240 and an MSE of 5492.4248. Visually, the model captured the general oscillatory behavior and approximate amplitude range, but its recursive predictions became less accurate over time due to error accumulation and phase drift. Overall, the LSTM is suitable for short-term prediction of this signal, but long-term recursive forecasting remains challenging.

REFERENCES

[1] E. N. Lorenz, "Deterministic nonperiodic flow," in *Universality in Chaos*, 2nd ed. London, U.K.: Routledge, 2017, pp. 367–378.